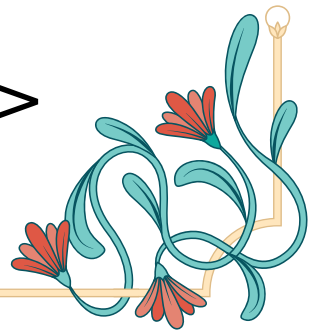


You're Measuring Memory Wrong: The Right Ways With DAMON

SJ Park <sj@kernel.org> <sjpark@crusoe.ai>



TL; DR

- Measure memory usage using DAMON
 - Enable Auto-tune
 - Record on labs, Snapshot on products
- Or, just ask DAMON to save memory
- Community is here to help

Table of Contents (40 mins)

- Background (5 mins)
- Tools (5 mins)
- Config (7 mins)
- Monitoring/Visualization (10 mins)
- One more thing (5 mins)
- QnA (8 mins)

Memory Efficiency: Expectation and Real

Quiz: What This Is?



Image source: https://media.licdn.com/dms/image/v2/D4D22AQEGuu4U82nddg/feedshare-shrink_800/B4DZ38iDPgHkAc-/0/1778058278141?e=2147483647&v=beta&t=1ZRbFrejThXU6Lqt7AGuBq_zwtJXhvlrDgazCp1Ep0

Quiz: What This Is?

- Hint: Not # of CVEs

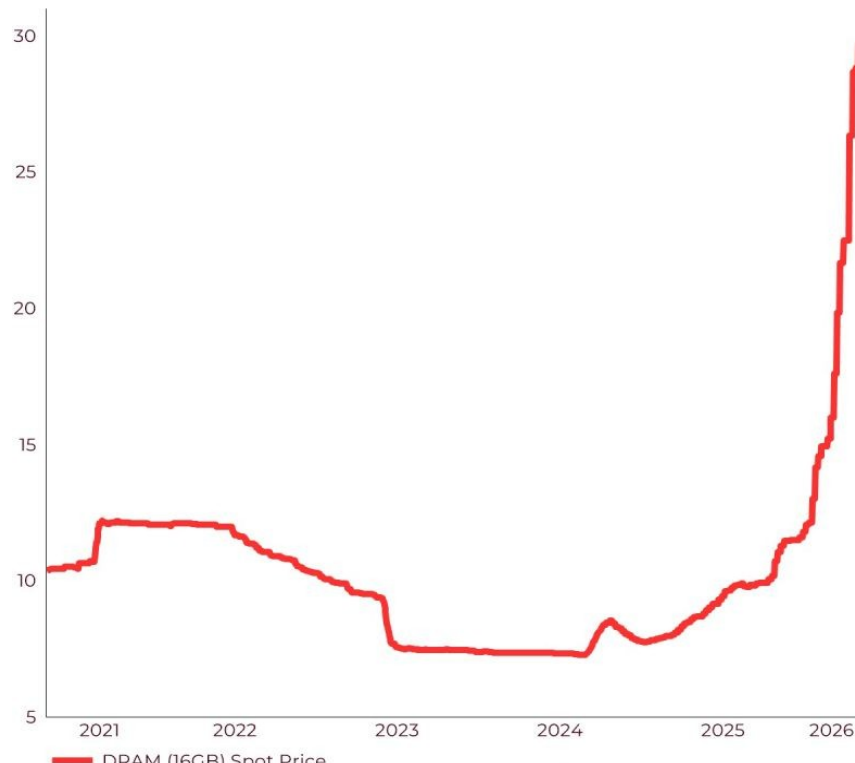
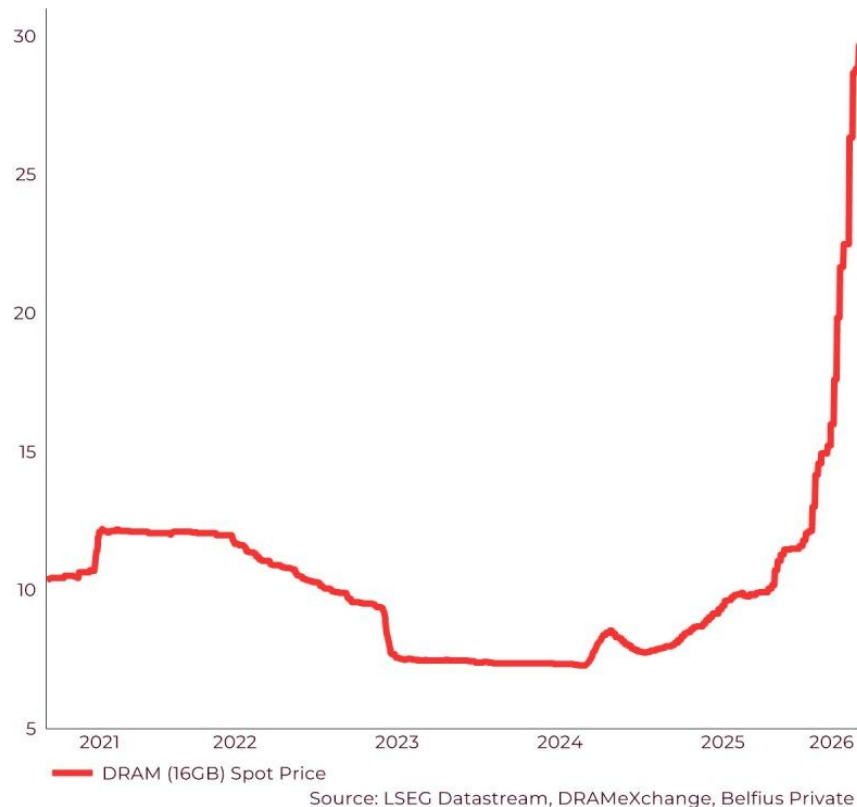


Image source: https://media.licdn.com/dms/image/v2/D4D22AQEGuu4U82nddg/feedshare-shrink_800/B4DZ38iDPgHkAc-/0/1778058278141?e=2147483647&v=beta&t=1ZRbFrejThXU6Lqt7AGuBq_zwtJXhvvzlrDgazCp1Ep0

Quiz: What This Is?

- Hint: Not # of CVEs
- Answer: DRAM price



Don't Panic, Overcommit

- Long time researched problem
- Equip RAM to only fit working set
- Kernel should do everything else

Thrashing, OOM Kills, Panic

- Having support != well supported
- Long time researched != Solved

“Everyone Has a Plan Until They Get Punched in the Mouth”,
Mike Tyson

Working Set

- Not necessarily all allocated memory
- Frequently accessed (hot) data
- How to monitor access?
- One of long-time unsolved issues

DAMON: Yet Another Solution

“DAMON **is** a Linux kernel subsystem for efficient data access monitoring and access-aware system operations.”

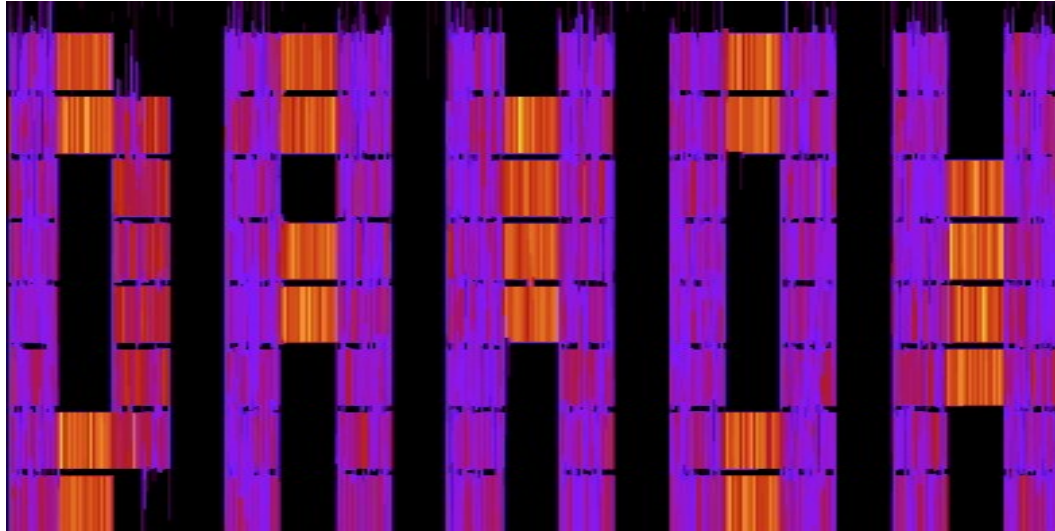


Image is **generated** by DAMON

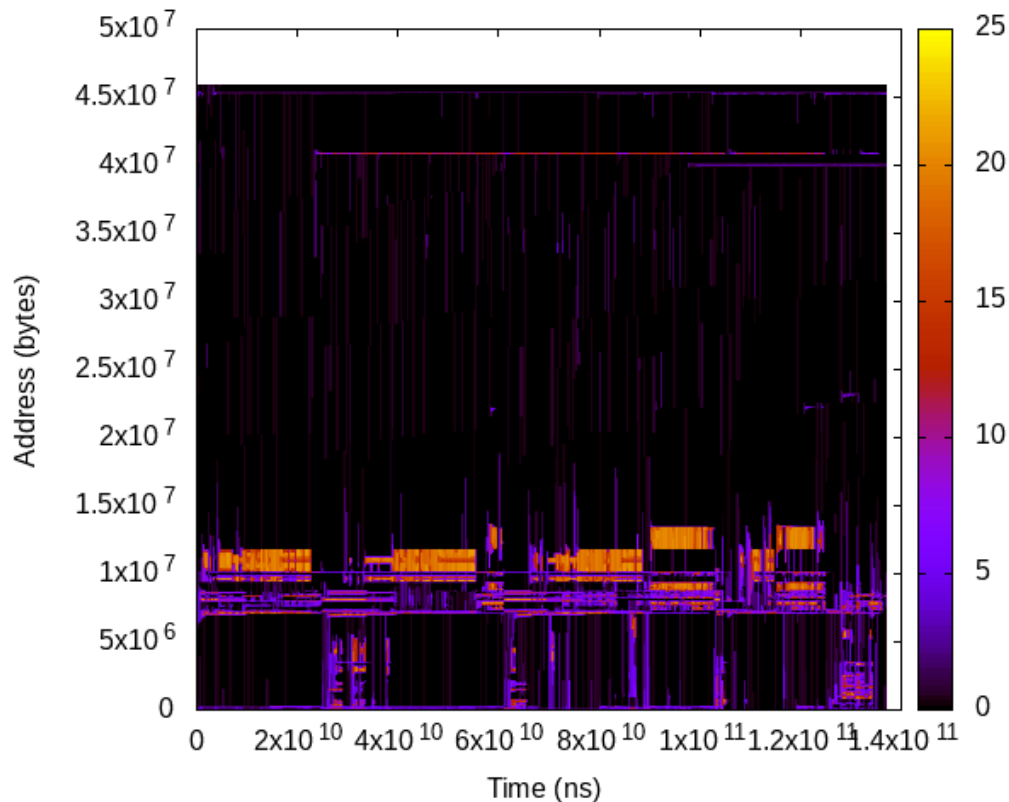
DAMON in Nutshell

High Level Implementation

- Spawn a kernel thread that does
- See if each page is accessed, every 5ms
- Inform users the findings, every 100ms
- Time intervals can be [auto-]tuned

Output: Access Information

- Location
- Frequency
- Stability
- Recency



Access pattern heatmap of
Splash2x/Raytrace

Design

- Goal: Accurate, light, tunable, automated
- Mechanisms: Sampling, Adaptive regions, Sysfs interface, Feedback loops
- Result: $<0.1\%$ CPU overhead on products
- Out of the scope of this talk

Availability and Usages

- Available on ≥ 5.15 upstream kernels
- Enabled on most [distro](#) kernels
- Being used for products and researches

Community

- Public channels
 - Mailing list: damon@lists.linux.dev
 - Project website: <https://damonitor.github.io/>
- Private channels
 - Maintainer email: sj@kernel.org
 - DAMON Beer/Coffee/Tea [Meetup](#)

Pain: Poor User Guides

- The evolution ain't be documentized
- Code changes faster than docs
- Users leave with misunderstandings
 - Don't be gentle...

The Guides Arrived

- This talk is the DAMON user guide
 - v2026.10.06; One version before another
- Ain't be complete; please report bugs
- Point is, placeholder for yelling

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How To Use

DAMON User-space Tool (damo)

- <https://github.com/damonitor/damo>
- Easy to use, flexible, not highly optimized

DAMON Sysfs Interface

- [/sys/kernel/mm/daemon/](#)
- Machine-friendly, flexible
- Optimized for “Build Your Own Tool”

DAMON Modules

- `CONFIG_DAMON_{STAT,RECLAIM,LRU_SORT}`
- Optimized for specific use cases
- More improvements (DAMON-X) will arrive

Tools: damo, sysfs, and AI

- Start from damo for simple usages
- Build your optimized tool using sysfs
- *AI could* make it easy (or worse)
 - Let AI refer to damo and avail resources
 - Request help to community if needed

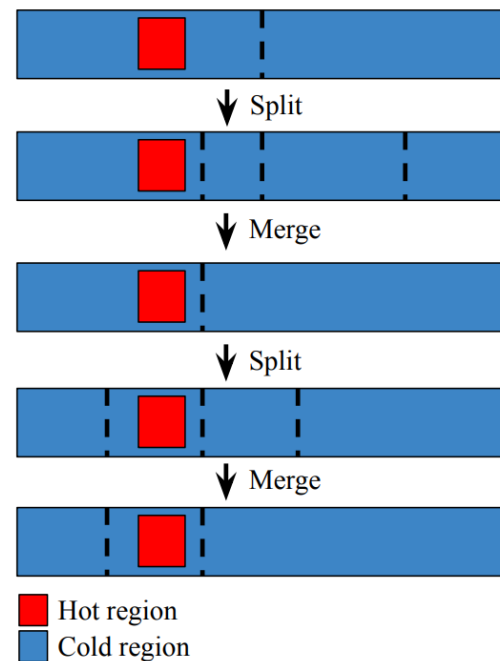
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DAMON Configuration

DAMON Region

- Address range of similar access frequency
 - Dynamic size, merge/split
- User sets # regions limits
 - Same to min/max accuracy/overhead



Regions Range Setup Guide

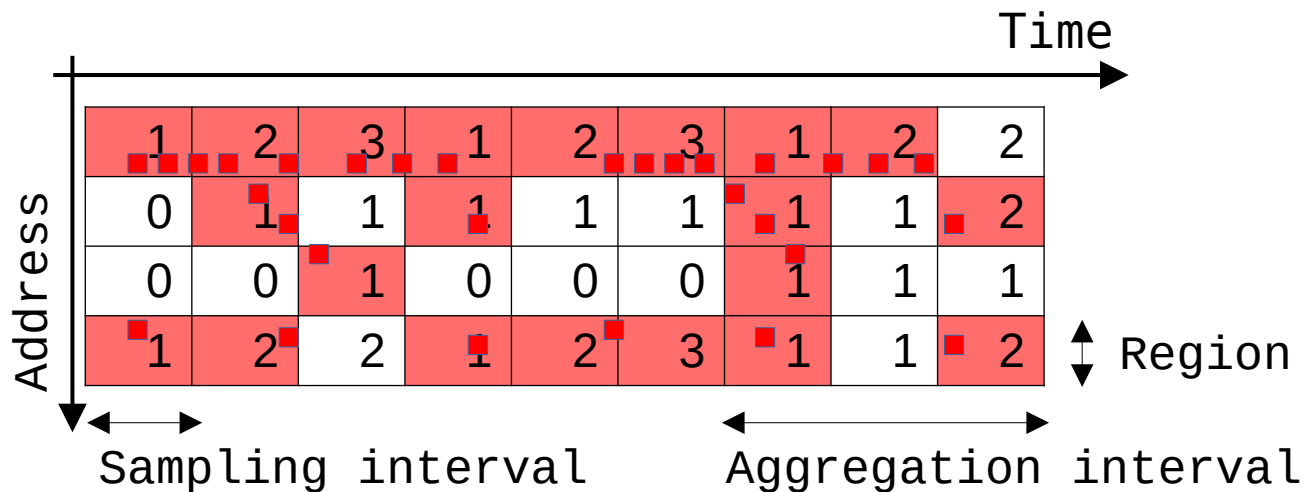
- Start from [10, 1000]
 - Proven to work on a few products
- Set min == max for fixed-gran monitoring

Monitoring Intervals

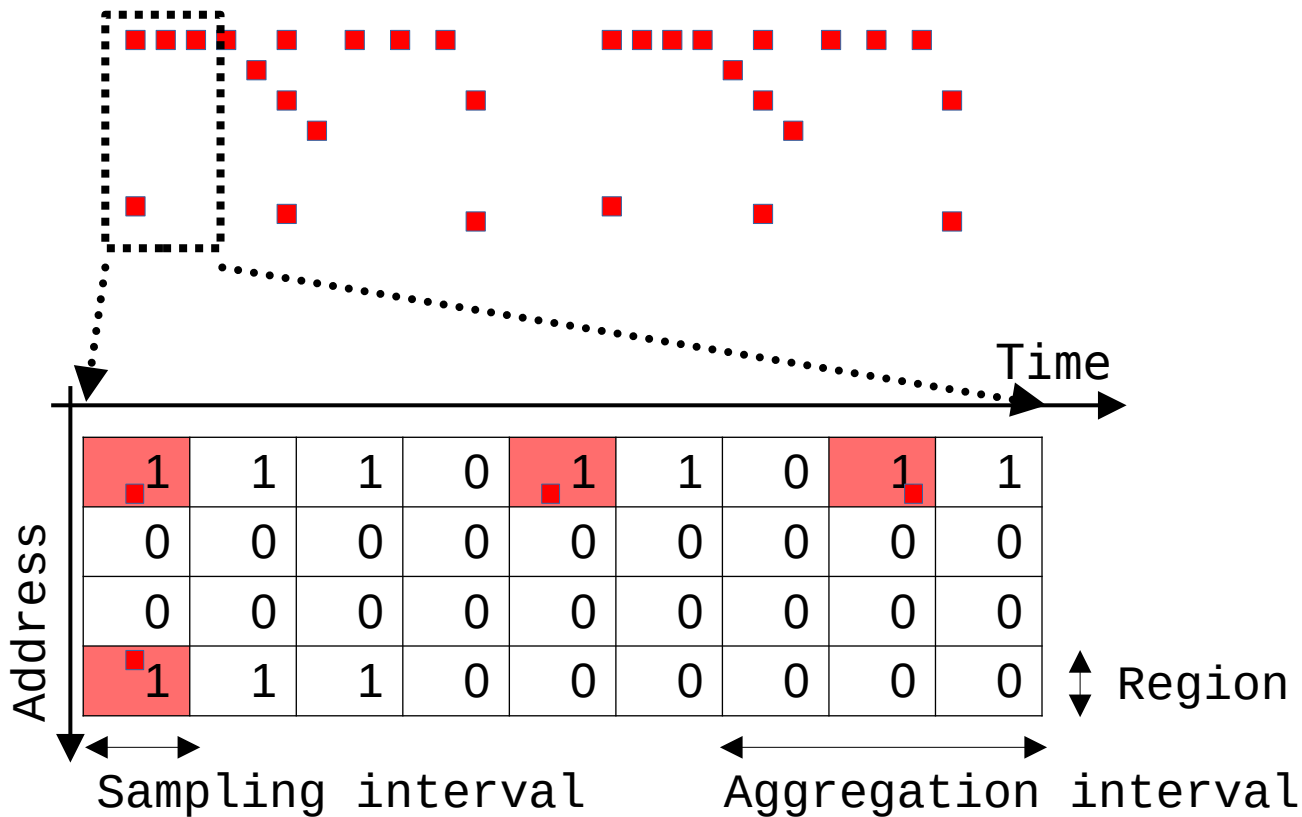
- Sampling interval (5ms by default)
 - Time between each page access check
- Aggregation interval (100ms by default)
 - Sampling results aggregation window
 - Time between monitoring results snapshot

Supposed DAMON Snapshot

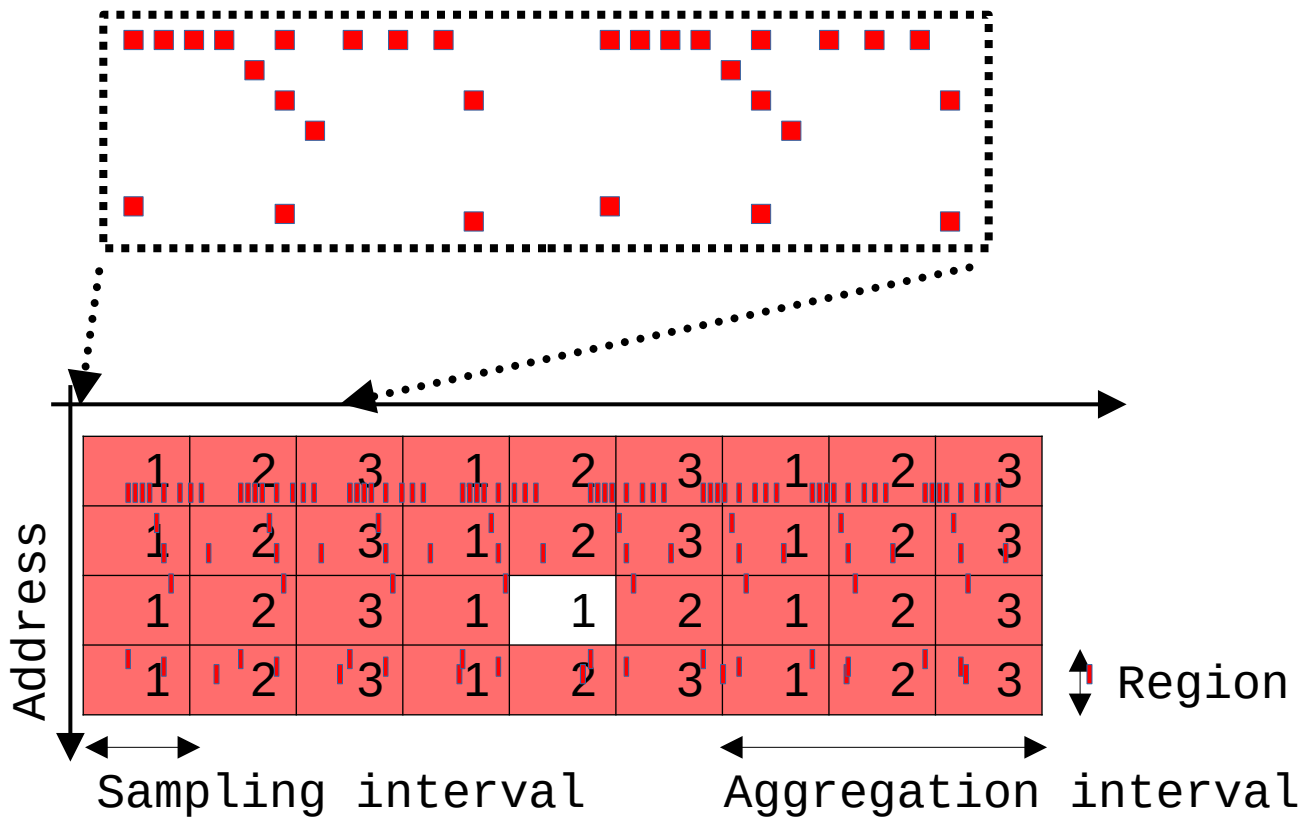
- Captures actionable amount of diversity



Short Intervals: Nothing Special



Long Intervals: All Special



Intervals Tuning Difficulty

- Nothing Special $\sim$ = All Special
 - Not actionable
- Optimum tuning depends on workload
 - Time consuming and repetitive
- DAMON provides intervals **auto-tuning**

Intervals Tuning Guide

- Start from auto-tuning
 - Aim 4% access event in each snapshot
 - Keep 1:20 sample:aggregation interval
 - Span sample interval in [5ms, 10s]
- Reach out to community for optimizations

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Capturing and Visualizing Monitoring Results

Monitoring Results Delivery

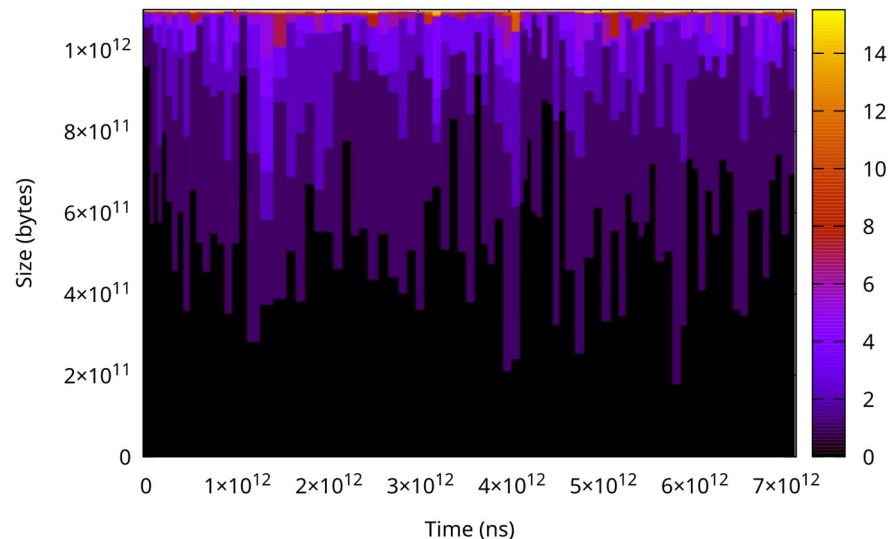
- Single snapshot built in kernel memory
- Exposed to user space via 2 ways
 - Tracepoint, every snapshot per aggregation
 - DAMON sysfs, one snapshot on demand

Record on Lab Envs

- Save all tracepoint output
- Interval auto-tuning doesn't matter
 - 5ms/100ms intervals can work
- Cost: space to save record
- Best for lab environments

Show Heatmap of Records

- Best for records
- Time-space zoom in/out is required
 - Otherwise, only a blue dot in the void



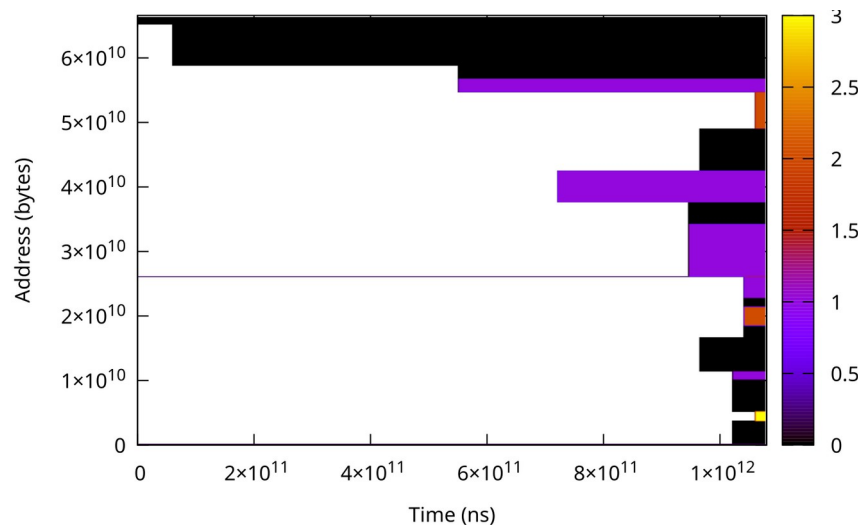
Heatmap shows ~750 GiB workingset

Record/Heatmap Demo

```
$ sudo damo record $(workload_pid)  
$ sudo damo report heatmap
```

Snapshot on Prod Envs

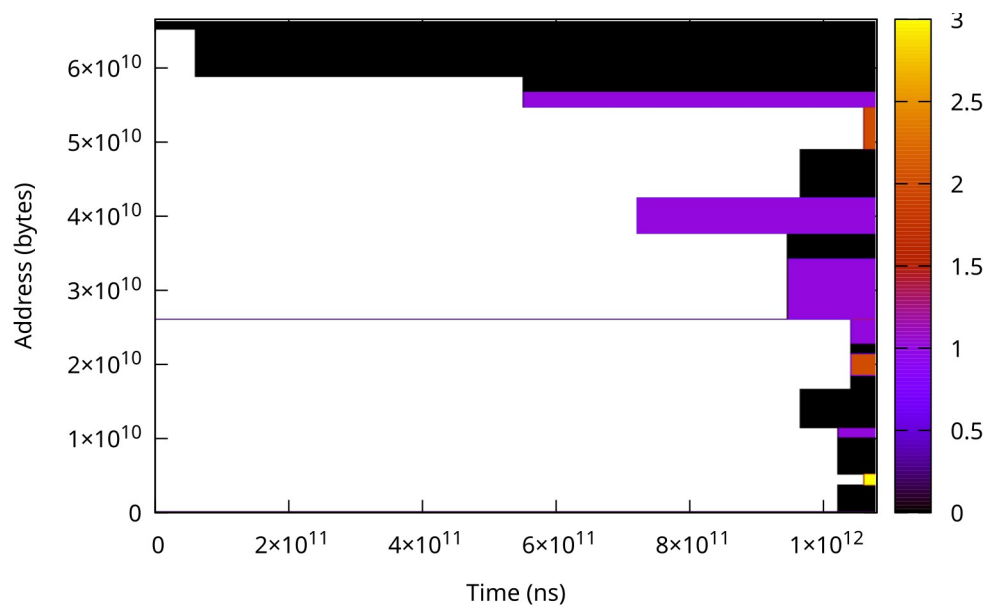
- Snapshot contains practical history
 - Access stability per region
- Light enough for prod envs: $<0.1\%$ CPU



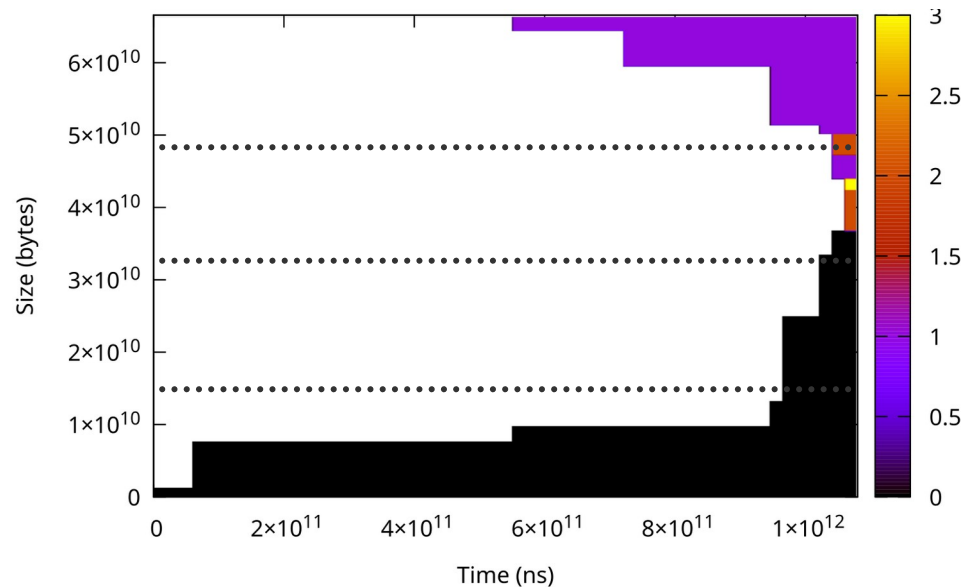
Idle Time Percentiles: Concept

- Idle time: how long it was not accessed
- Percentile: that of the statistics

Unsorted snapshot



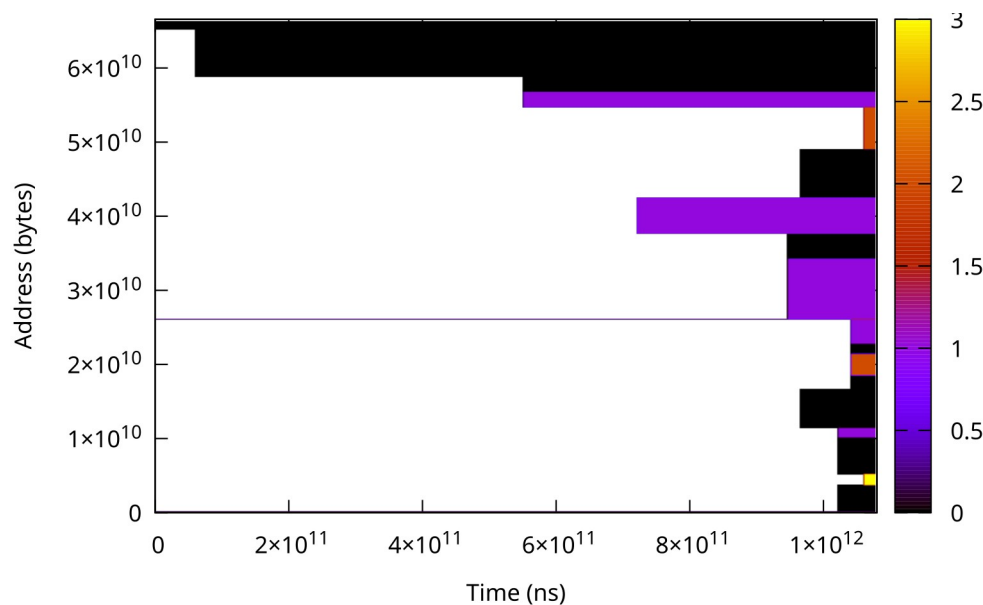
Sorted by access frequency



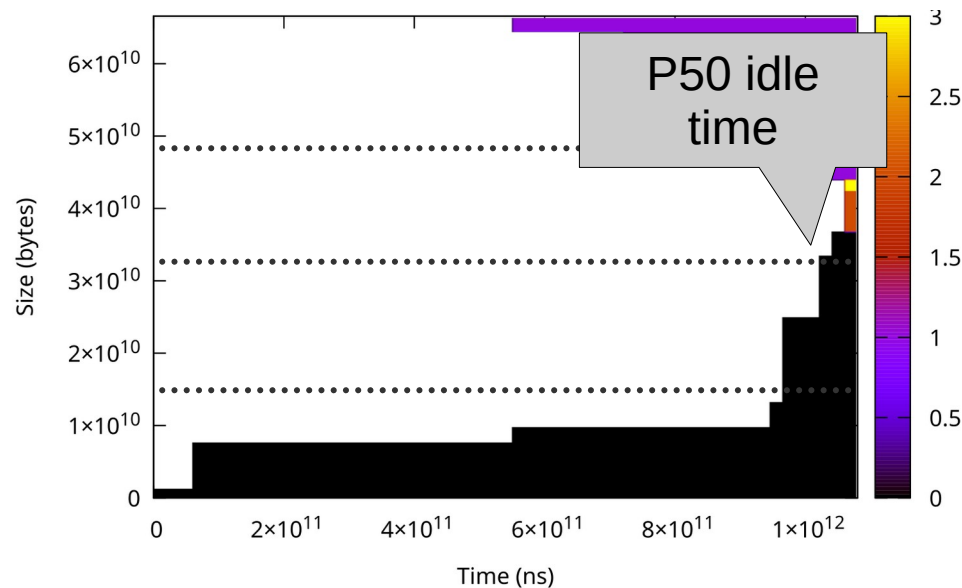
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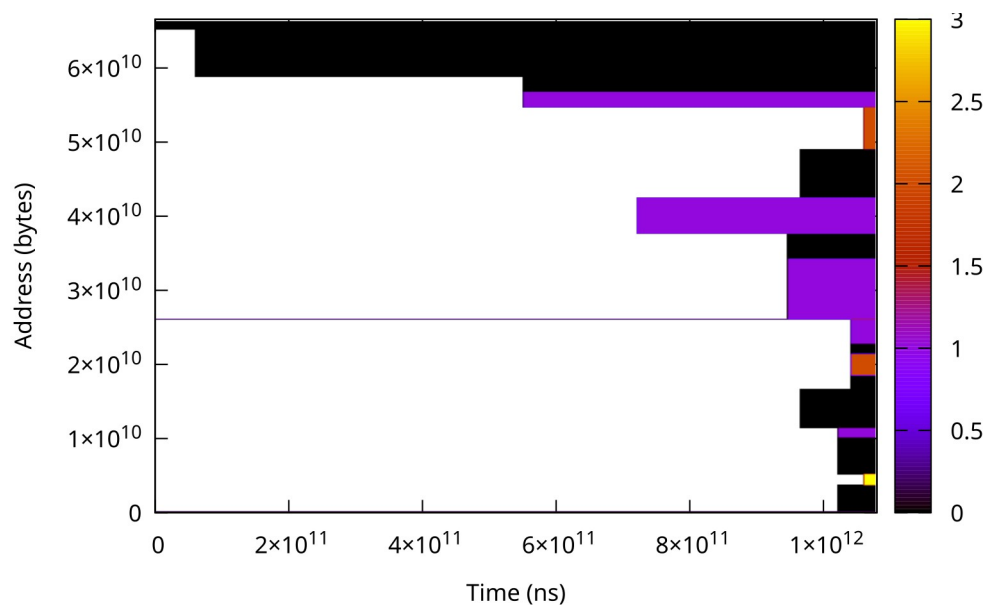
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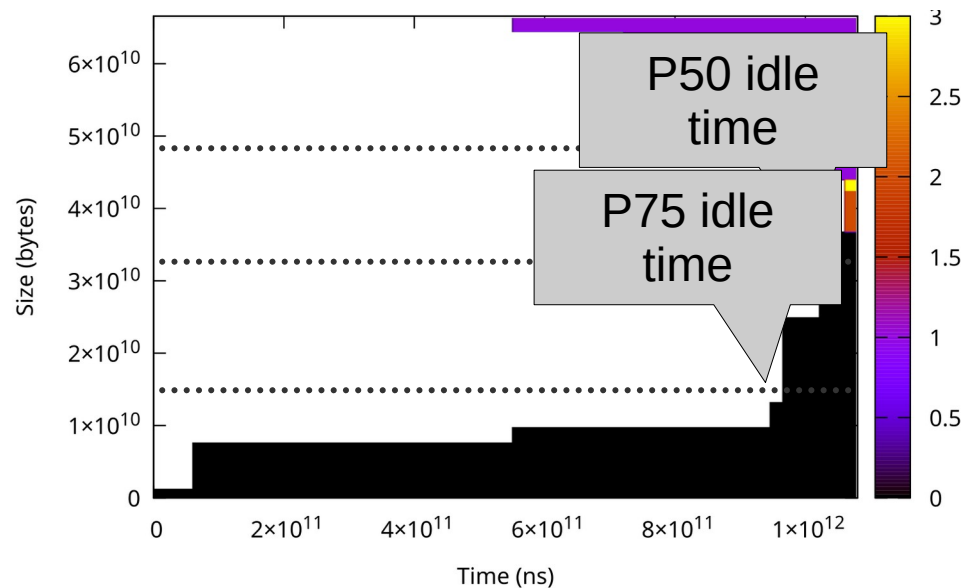
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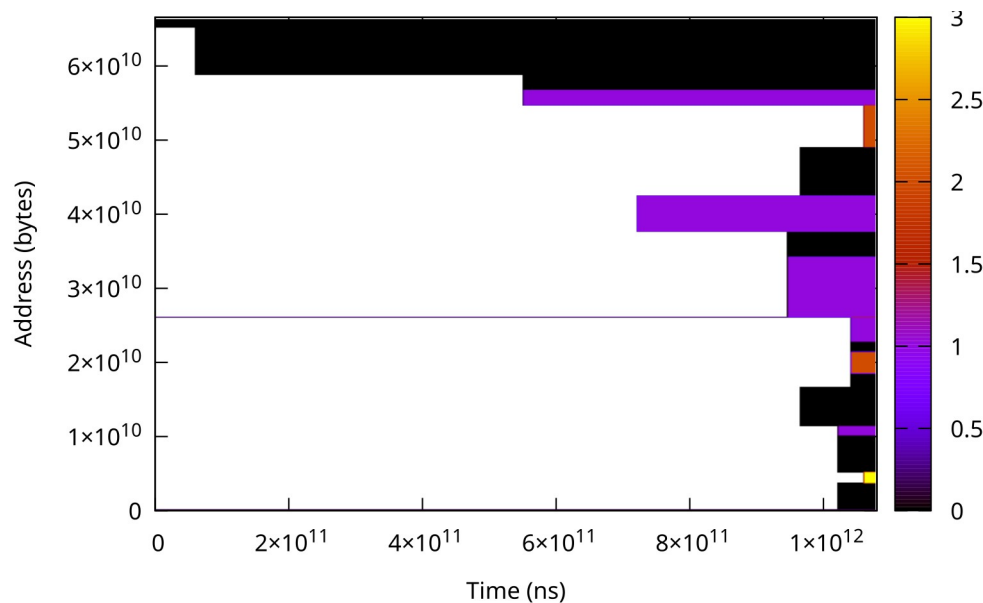
Sorted by access frequency



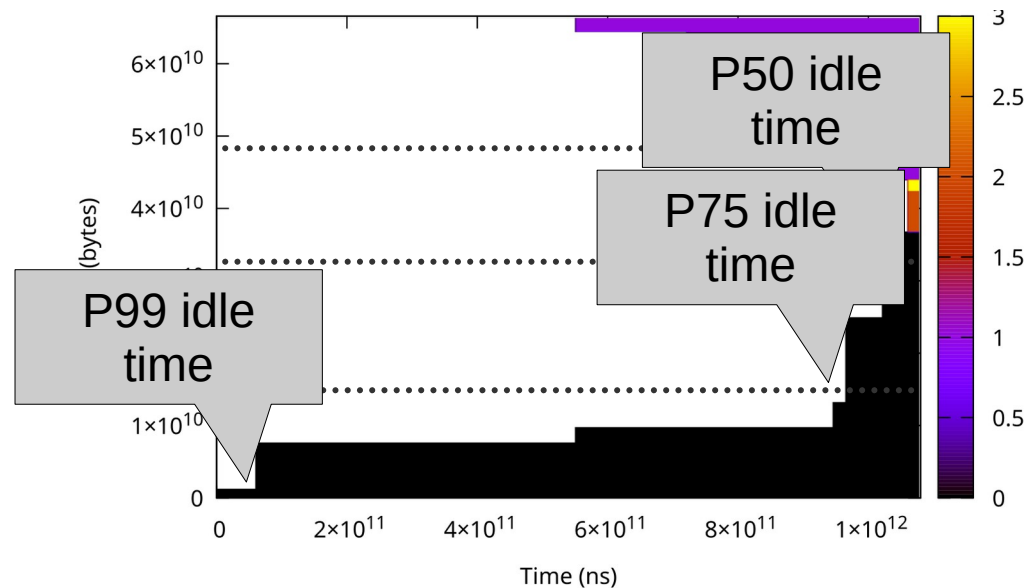
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- Percentile: that of the statistics

Unsorted snapshot



Sorted by access frequency

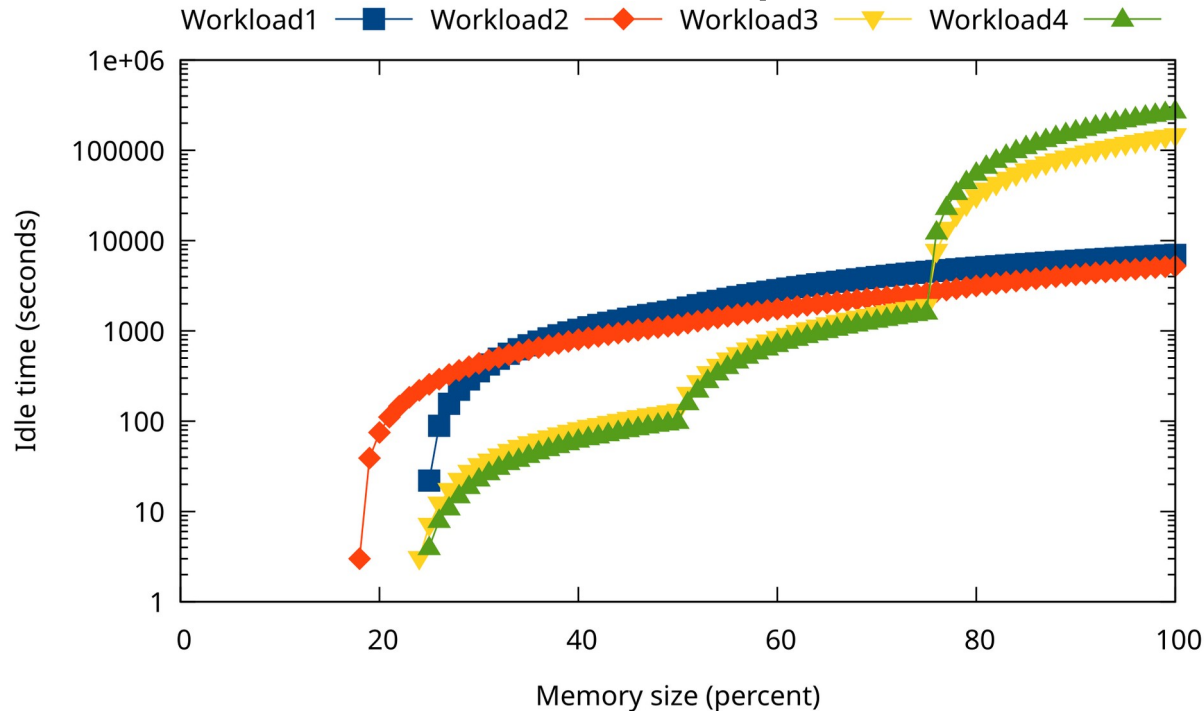


Idle Times for Snapshot

- Easy to aggregate fleet-wide snapshots
 - E.g., Fleet-average $p\{25,50,75\}$ idle time
- Integrate well into Grafana-like systems

Idle Time Percentiles in Real

- Note: Y-axis in logscale
- Show common and different patterns of workloads



Snapshot/Idle Times Demo

```
$ sudo damo start $(workload_pid)  
$ sudo ./damo report access --style idle-time-percentiles
```

Memory Monitoring Wrapup

- Config DAMON as community suggests
- Record and show heatmap on lab
- Snapshot and show idle time on prod/fleet

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One More Thing: DAMOS

Monitoring: Only One of Issues

- How hot data is workingset?
- How kernel finds the workingset?
 - User sends hint to kernel?
- DAMON is a part of kernel; Can't it do all?

DAMOS: Second Face of DAMON

- DAMon-based Operation Schemes
 - “Page out cold memory”
 - “Migrate hot/cold pages to DRAM/CXL node”
- Turn DAMON into access-aware memory management system operations engine

DAMOS First

- DRAM price is already increased
- No time to observe first
- Can we save memory first?
- Running proven DAMOS schemes is a way

Proactive Reclaim

- Proactively reclaim cold memory
- Users: Google, Meta, AWS, DeepSeek
 - Commonly reports ~20% memory saving
 - AWS and DeepSeek uses DAMOS-based one
- One well proven overcommit approach

DAMOS for Proactive Reclaim

```
$ sudo ./damo start --monitoring_intervals_autotune \  
    --damos_action pageout --damos_access_rate 0% 0% \  
    --damos_age 2m max \  
    --damos_quota_space 128M --damos_quota_interval 1s \  
    --damos_quota_goal some_mem_psi_us 5000
```

DAMOS for Proactive Reclaim

Hey damo, start DAMON for monitoring
entire physical address space

```
$ sudo ./damo start --monitoring_intervals_autotune \  
--damos_action pageout --damos_access_rate 0% 0% \  
--damos_age 2m max \  
--damos_quota_space 128M --damos_quota_interval 1s \  
--damos_quota_goal some_mem_psi_us 5000
```

DAMOS for Proactive Reclaim

I don't know optimum sampling/aggregation intervals,
so just auto-tune the intervals.

```
$ sudo ./damo start --monitoring_intervals_autotune \  
--damos_action pageout --damos_access_rate 0% 0% \  
--damos_age 2m max \  
--damos_quota_space 128M --damos_quota_interval 1s \  
--damos_quota_goal some_mem_psi_us 5000
```

DAMOS for Proactive Reclaim

Pageout memory that not accessed for ≥ 2 minutes

```
$ sudo ./damo start --working_intervals_autotune \
--damos_action pageout --damos_access_rate 0% 0% \
--damos_age 2m max \
--damos_quota_space 128M --damos_quota_interval 1s \
--damos_quota_goal some_mem_psi_us 5000
```

DAMOS for Proactive Reclaim

```
$ sudo .
```

But only up to 128 MiB/second. I don't want too rapid reclaim.

```
--damos_access_rate 0% 0% \
--damos_age 2m max \
--damos_quota_space 128M --damos_quota_interval 1s \
--damos_quota_goal some_mem_psi_us 5000
```

DAMOS for Proactive Reclaim

```
$ sudo .
```

Under the hard speed limit, adjust the real speed
aiming 5ms/second (0.05%) memory pressure

```
--damos_quota_space 128M --damos_quota_interval 1s \  
--damos_quota_goal some_mem_psi_us 5000
```

Just Run Proactive Reclaim

- Test first, of course
- It should save $\sim 20\%$ memory for you
- If not, the community is there to help

Disclaimer

No Satisfaction Guaranteed

- Only recommendation
- Test and deploy under your charge

Community Exists for Reasons

- Place to report issues and cowork
- Possible services
 - Advanced tuning guides
 - Investigations
 - Feature backports

Wrapup

- Don't panic, use DAMON
- Start from recommended configs
- Measure real memory usage
 - Recording/heatmap on lab
 - Snapshot/idle time percentiles on prod
- Start from DAMOS if it works

QnA

- Feel free to use
 - Hallway track
 - damon@lists.linux.dev
 - DAMON Beer/Coffee/Tea [Chat](#)
 - sj@kernel.org